

A Candidate Astronomical Configuration for the Kali Yuga Epoch: Retrocalculation and Historical Transmission

Michael Taktikos

Karsten Müller

September 2026

Abstract

For over two centuries, Western Indology and the history of science have struggled to reconcile the traditional Indian epoch of the Kali Yuga (KY) with modern astronomical retrocalculations. Some scholars have argued that the traditional epoch is a statistically flawed projection derived from medieval observations [2]. This paper presents a different approach. We take as our starting point the widely discussed idea of a planetary configuration at the KY epoch. We do not define the configuration too narrowly, so as not to filter out alternative astronomical possibilities. We therefore adopt a generous upper bound: a planetary concentration distributed across two zodiacal regions. This is not intended as a reinterpretation of the KY sources, but only as the upper limit of our candidate search. We then use the modern planetarium program Stellarium[23] for visual verification. For the purposes of this study, a candidate passes this visual check if Stellarium displays all five naked-eye planets simultaneously in the selected sky view. Because we seek candidates throughout the millennium from 3500 BCE to 2501 BCE, manually entering all 365,000 days into Stellarium is impractical. We therefore locate possible candidates with a heuristic Python program using Skyfield[25] and the JPL DE441 ephemeris[24], together with two independent heuristic Mathematica[22] programs. All three approaches identify the same two-day window, 17–18 February 3103 BCE (proleptic Julian), during which Mercury, Venus, Mars, Jupiter, and Saturn form a broad planetary concentration near ζ Piscium (Revatī). We compare this candidate with the widely cited dates 18 Feb 3102 BCE and 18 Feb 3101 BCE. The visual test of all three dates in Stellarium shows that the 3103 BCE window satisfies the adopted graphical criterion, whereas the two alternative dates examined do not. We further propose two secondary historical hypotheses: that the 3102 BCE date may reflect confusion between historical BCE notation and Jacques Cassini’s astronomical year numbering [4], and that the 3101 BCE date may reflect a sign error in converting an astronomical integer to a historical year. These historical hypotheses are speculative and require archival confirmation. The astronomical result is a candidate, not a proof, and depends on the criteria used to define a planetary concentration.

1 Introduction: Two Common Dates for the Kali Yuga

The present study began from a chronological puzzle. Two dates are commonly cited for the onset of the Kali Yuga: 18 February 3102 BCE (proleptic Julian) and 18 February 3101 BCE, the latter associated with David Pingree. The two differ by exactly one year.

The earlier date is often attributed, anachronistically, to Indian astronomers such as Āryabhaṭa in the fifth century CE. However, the CE/BCE year-numbering system was introduced by Dionysius Exiguus only in the sixth century, so such a date format cannot have originated with Āryabhaṭa. A search for the first explicit publication of the date in BCE form yields no earlier source than John Playfair’s “Remarks on the Astronomy of the Brahmins” (1790) [13]. Neither Playfair nor Pingree disclosed a complete algorithm for arriving at their dates. The question therefore arises: how were these dates obtained, and why do they differ by exactly one year?

2 Textual and Historiographical Background

The search for the KY epoch is complicated by a plurality of narrative traditions. The epoch has been linked to a solar eclipse, a flood, and a planetary concentration. Textual criticism and improved translation have clarified many passages, but they cannot by themselves determine whether any of these narratives describes a real astronomical event.

Western academic literature commonly offers 18 Feb 3102 BCE or 18 Feb 3101 BCE as possible KY epochs. Planetarium software shows neither a solar eclipse nor a strict or partial conjunction of all naked-eye planets on those dates.

The source situation surrounding Āryabhaṭa is also difficult. There are no extant autograph sources, and the statements attributed to him are sometimes contradictory. O'Connor and Robertson note errors in fundamental formulas that an astronomer would hardly have made; Ganitānand calls these "mathematical lapses." Elfering, by contrast, attributes such problems to translation errors. The unsatisfactory state of the evidence has even led some modern critics to hypothesize two distinct figures named Āryabhaṭa.

A widely repeated numerical riddle concerns the 3,600th year of the KY. According to one tradition, Āryabhaṭa had completed twenty-three years at that time, and his birth year is considered "established" as 476 CE, from which 3102 BCE can be calculated as the KY epoch. The English Wikipedia entry on Āryabhaṭa, however, takes 3102 BCE as its starting point and deduces that 499 CE marked his twenty-third year, implying a birth year of 476 CE. The argument is circular: 3102 BCE implies 476 CE, and 476 CE implies 3102 BCE.

Roger Billard [2] applied a least-squares statistical analysis and concluded that the planetary parameters of the *Āryabhaṭīya* fit best around 499 CE. He suggested that the KY epoch was a theoretical projection backward from fifth-century observations [9]. However, the present Skyfield search does not identify a comparable planetary concentration, with or without a solar eclipse, in the fifth or sixth century. Besides Billard, another critic of early Indian astronomy was the aforementioned Pingree, a proponent of a paradigm that, for decades, treated systematic astronomy in early India as substantially less developed than later Greek and Islamic traditions [12].

The publications of Subhash Kak challenged this view. Kak demonstrated that architectural and textual layers of the *Rigveda* contain a sophisticated code tracking the synodic periods of planets [18]. This rekindled our interest in the problem and led us to examine a broader range of sources, including Arabic texts such as those of Abu Maʿshar, whose works preserve translations of Indian originals made at the House of Wisdom.

The most widespread astronomical interpretation is based on planetary concentration, but it should be explicitly noted that it is not the only one. For example, some translators like Dr S. Balakrishnan and K. S. Raghavan interpret mentions of two dark periods in the Mahabharata as a solar eclipse followed after 13 days by a lunar eclipse [14], which narrows down the possible occurrence of these events considerably. While this interpretation is incompatible with planetary concentrations, one can give absolute priority to the solar eclipse interpretation and disregard all other texts mentioning planets. A scientist who took the pure-eclipse approach but challenged the translation of Raghavan and included a third eclipse is Narahari Achar, who calculated the astronomical events in his interpretation with the program SkyMap Pro and concluded that the Mahabharata war happened 22 Nov 3067 BCE [15].

In this work, we focus on what date would result if, conversely, the planetary concentration interpretation were given priority and any potentially concurrent dark skies did not constitute an eclipse. As noted above, Skyfield shows that the planetary constellation could not have been a strict or even partial conjunction of all planets. This would invalidate interpretations of the *Sūryasiddhānta* according to which, at the end of the Kṛta-yuga, the mean planets were aligned at the beginning of Aries. Other historians follow Brahmagupta's tradition and associate the reference point with Revatī. "Revatī" may be interpreted as a lunar mansion extending over

a region of the ecliptic, not necessarily identical to a single physical star. The astronomical tradition also associates "Revatī" with a junction star, commonly identified with ζ Piscium. In the first interpretation, a statement that the planets occupied the "end of Revatī" need not mean that they were all within a small angular distance of ζ Piscium.

According to al-Maqrīzī's account of Abū Maʿshar [8], the planetary concentration was linked to the Great Flood and was described as a broader configuration spanning the end of the sign of Pisces and the beginning of the sign of Aries:

"وقال: كان الطوفان عند اجتماع الكواكب في آخر برج الحوت، وأول برج الحمل"

Translation: "And he said: The Great Flood occurred at the conjunction of the planets at the end of the sign of Pisces and the beginning of the sign of Aries."

Terms such as "true conjunction" and "partial conjunction" are modern astronomical terms that had not been introduced in the lifetime of Āryabhaṭa or Abu Maʿshar. They are useful today, but they may impose an overly restrictive filter on ancient descriptions. We therefore do not attempt to introduce a new interpretation of the KY sources. Instead, we use a generous search criterion: a concentration distributed across two zodiacal regions rather than a tightly clustered conjunction. The criterion is intended to avoid filtering out possible candidates. We do not require the planets to have been visible from India; rather, we require that the configuration be calculable from Ujjain and visually demonstrable in Stellarium as a topocentric configuration.

3 Astronomical Method and Results

To test the historical interpretation discussed by Kak, we searched the entire millennium from 3500 BCE to 2501 BCE. Manually entering 365,000 dates into Stellarium and inspecting the resulting sky views is not practical.

We therefore implemented a heuristic daily scanning algorithm in Python using Skyfield and JPL DE441. We also wrote two independent algorithms in Wolfram Mathematica. All three programs calculate ecliptic-longitude differences for Mercury, Venus, Mars, Jupiter, and Saturn relative to one another and to the junction star ζ Piscium (01) over the full millennium from 3500 BCE to 2501 BCE. The observational point is Ujjain, India. The criteria adopted are heuristic: Venus–Mercury separation must be less than 3° ; Mercury, Venus, and Mars must lie within 7° of Revatī; and the outer planets may span up to 60° , corresponding to two zodiac signs. These thresholds are intended to capture a broad concentration rather than a strict conjunction.

Remarkably, all three algorithms isolate the same two-day window:

1. 17 February 3103 BCE
2. 18 February 3103 BCE

We do not claim that Āryabhaṭa associated the KY epoch with a planetary configuration with exactly this particular heuristic calculation in mind. Our calculation is only a search aid for determining which dates to inspect in Stellarium. The question is whether this time window contains a candidate under the deliberately broad criteria described above. To check this, we used Stellarium to display the sky over Ujjain between midnight and sunrise on 18 February 3103 BCE. Although the display shows all five naked-eye planets simultaneously and confirms that a rare planetary concentration took indeed place on this date, the selected sky view also reveals that the planets were not above the horizon to be simultaneously visible to an observer at Ujjain; the only place where they were above the horizon was Antarctica, and February fell in the 6 months where there was daylight in the Antarctic, so not even there the planets could have been simultaneously visible to an observer. This establishes that the planetary concentration on 18 February 3103 BCE was retrocalculated by Āryabhaṭa. Only

this date satisfies in the Stellarium program our graphical criterion for the required topocentric configuration, the configuration appears in the selected Stellarium sky view. The same criterion is not met on 18 February 3102 BCE or 18 February 3101 BCE.

The traditional Indian chronology assumes that an extraordinary planetary configuration occurred at the KY epoch. Our retrocalculation and the Stellarium graphics identify a planetary configuration compatible with one historically attested interpretation of "Revati" and the KY. Implementations of alternative configurations and interpretations—such as solar eclipses combined with planetary concentrations—did not find real-world events. We stress that the KY epoch can be defined in other ways, including ways that involve impossible astronomical constellations and need not have real-world manifestations. Our claim is narrower: one real-world astronomical event exists that is consistent with a traditional planetary KY concentration spanning two constellations.

In our Skyfield program, relaxing the inner-planet thresholds slightly produces an additional date, 19 February 3103 BCE. We leave a systematic sensitivity analysis of how the number of candidates changes as the thresholds are relaxed for a future study. In the present paper, we test the candidate 18 February 3103 BCE directly against the two traditionally transmitted dates, 18 February 3102 BCE and 18 February 3101 BCE. We could not relax the conditions while retaining the minimum requirement that the outer-planet separations remain within a 60° ecliptic-longitude span in such a way that either 18 February 3102 BCE or 18 February 3101 BCE would appear.

4 Comparison with the cited Alternative Dates

We modified the Python program to output the ecliptic-longitude differences for the candidate date 18 February 3103 BCE and for the alternative dates 18 February 3102 BCE and 18 February 3101 BCE, and we also entered the alternative dates in Stellarium for visual comparison. The results are shown in Table 1. Under the adopted heuristic criterion, the 3103 BCE dates satisfy the required concentration; the two alternative dates do not.

Date	V–M	Ma–M	Ma–V	J–M	J–V	J–Ma	S–M	S–V	S–Ma	J–S
3103 BCE 17 Feb	2.66	3.97	6.64	42.13	44.80	38.16	53.47	56.13	49.49	11.33
3103 BCE 18 Feb	2.21	4.91	7.12	43.64	45.85	38.73	55.08	57.29	50.17	11.44
3102 BCE 17 Feb	4.42	102.84	107.26	28.00	23.58	130.84	56.53	52.11	159.37	28.53
3102 BCE 18 Feb	7.08	101.10	108.18	29.89	22.81	130.99	58.57	51.49	159.66	28.67
3101 BCE 17 Feb	6.79	22.09	28.88	18.30	25.09	3.79	62.61	69.40	40.52	44.30
3101 BCE 18 Feb	6.05	23.33	29.38	20.03	26.08	3.30	64.47	70.52	41.14	44.44

Table 1: Ecliptic-longitude differences in degrees. V = Venus, M = Mercury, Ma = Mars, J = Jupiter, S = Saturn. Under the adopted heuristic two-region criterion, only the 3103 BCE dates satisfy the concentration.

Alternative hypothesis A (18 February 3102 BCE) does not satisfy the adopted criterion because the separations of Mars–Venus, Jupiter–Mars, and Saturn–Mars exceed 100° , placing those pairs more than two zodiacal regions apart. The Saturn–Mars separation is 159.37° . Alternative hypothesis B (18 February 3101 BCE) does not satisfy the adopted criterion because Saturn–Mercury and Saturn–Venus exceed the two-region tolerance of 60° .

Entering all three dates in Stellarium requires some care because Stellarium uses astronomical year numbering; for example, -3102 corresponds to 3103 BCE. Stellarium also displays the weekday name alongside the numerical year, which provides a useful check on the date conversion. With English language settings, the output dates would be: Thu, Feb 18, -3102; Fri, Feb 18, -3101; and Fri, Feb 18, -3100. With our preferred German-language settings, they are: Do, 18 Feb -3102; Fr, 18 Feb -3101; Fr, 18 Feb -3100.

The resulting views are shown below. For both 18 February 3102 BCE and 18 February 3101 BCE, the Stellarium graphics do not contain all five naked-eye planets in the selected sky view, unlike the 18 February 3103 BCE view.

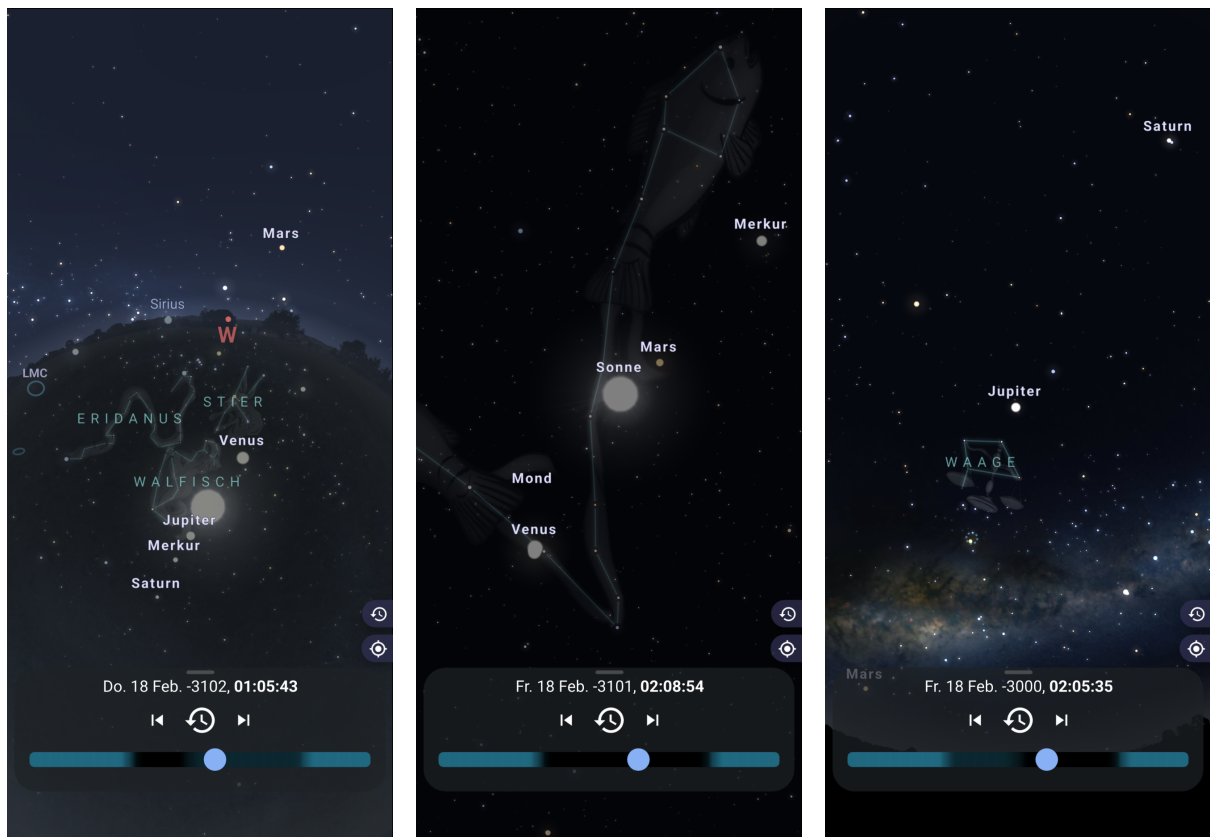


Figure 1: Ujjain, 18 February 3103 BCE.

Figure 2: Ujjain, 18 February 3102 BCE.

Figure 3: Ujjain, 18 February 3101 BCE.

Figure 4: Stellarium views over Ujjain at approximately 1:00 a.m. local time on 18 February for the years 3103 BCE, 3102 BCE, and 3101 BCE.

The two counter-hypotheses therefore do not meet the minimum requirement on the dates examined.

When the candidate date of 18 February 3103 BCE is converted into the classical Hindu solar calendar via Mathematica, the system yields:

Brihaspatvara, 1 Mesha -1 K.Y.

where *Brihaspatvara* translates as **Thursday**.

A chronological complication arises from the precise time of day attributed to the epoch's inception. Traditional Puranic accounts state that Krishna departed the earth in the evening or night hours. In ancient calendar systems, the boundary defining the transition from one weekday to the next was school-dependent:

1. **The Audayika system (sunrise):** Civil and ritual days begin at local sunrise. An event occurring after sunset remains under the domain of the current daytime integer.
2. **The Ardharatrika system (midnight):** Mathematical and astronomical tables shift the calendar date at midnight, aligning with Western proleptic calendars.

The debate over whether the astronomical weekday of the KY epoch should be assigned to Thursday or to the beginning of the new day (Friday) had not been resolved as late as the ninth century, in the time of Abū Maʿshar. The historical tradition of Āryabhaṭa does not define the day as beginning at midnight but at sunrise. Al-Maqrīzī clarifies:

"ومنهم من يرى أن الطوفان كان يوم الجمعة، وعند أبي معشر أنه كان يوم الخميس"

"Some held the view that the Flood occurred on a Friday; according to Abū Maʿshar, it took place on a Thursday."

For Abū Maʿshar, the day of the week and the astronomical epoch were factors in the calculation.

5 Two Secondary Historical Hypotheses

The exact annual intervals between our candidate date and the two commonly cited dates—one year from Playfair’s 18 February 3102 BCE and two years from Pingree’s 18 February 3101 BCE—motivate two secondary hypotheses. We emphasize that these are historical conjectures rather than established explanations.

5.1 Secondary Hypothesis A: A Mistake in Playfair’s 3102 BCE

In 1740, Jacques Cassini introduced astronomical year numbering to streamline ephemeris calculations [4]. This system introduced the year 0, so that historical years Before Common Era are shifted by one unit when expressed as mathematical integers:

$$\text{Year } X \text{ BCE} = -(X - 1). \quad (1)$$

Thus, the historical year 3103 BCE is written in astronomical tables as -3102 .

Not long after Cassini’s innovation, Guillaume Le Gentil [5] brought Indian astronomical tables to Europe, including the Tirvalore Tables, which did not contain a "BCE" marker but rather a continuous year count. Pioneering European researchers such as John Playfair [13] and Jean-Sylvain Bailly [1] then published the date 18 Feb 3102 BCE, exactly one year away from the planetary concentration revealed by the Skyfield data.

Our hypothesis is that one of these researchers cross-referenced Indian astronomical tables with contemporary European ephemerides, mapped the relevant alignment data to Western tables, obtained the astronomical year integer -3102 , failed to account for the year-zero convention, and then rendered that integer as 3102 BCE. This proposed chain of transmission is conjectural and requires archival confirmation; the present evidence does not establish that Playfair or Bailly actually followed this procedure.

5.2 Secondary Hypothesis B: A Mistake in Pingree’s 18 Feb 3101 BCE

David Pingree associated the theoretical epoch with 18 February 3101 BCE [12]. This date is a Friday, and the present calculations do not identify an astronomical concentration meeting the adopted criteria on that day; the corresponding Stellarium view likewise does not contain all five naked-eye planets in the selected sky view.

Our hypothesis is that Pingree correctly recognized that the traditional text-based tables implied an astronomical zero-point located at integer -3102 , but attempting to convert this back

into a proleptic Julian historical year, he inverted the directional shift of negative time. Instead of correctly subtracting a year to account for the absent historical year zero, he mistakenly added a year, producing 3101 BCE. This would explain the two-year distance between Pingree’s date and the date revealed by the Skyfield data. Again, this hypothesis is speculative and requires textual and archival evidence.

Taken together, these hypotheses could explain the transmission of the dates 18 February 3102 BCE and 18 February 3101 BCE as consequences of errors in converting between astronomical and historical dating systems. At present, however, they remain unverified conjectures, and alternative explanations remain possible.

6 Discussion and Limitations

Several limitations should be borne in mind. The astronomical criteria used to find the dates are heuristic. The thresholds 3° , 7° , and 60° are not directly prescribed by the ancient texts. The visual analysis helps determine whether the 3103 BCE configuration is reproduced in the selected Stellarium sky view. That view includes all five naked-eye planets for only one of the three checked candidates, namely 18 February 3103 BCE. The calendar conventions require careful standardization. Stellarium and Skyfield use astronomical year numbering, whereas historical BCE notation differs by one unit because historical BCE notation has no year zero. To avoid confusion about whether the dates are expressed in the proleptic Julian calendar, the proleptic Gregorian calendar, or astronomical year numbering, we state the convention explicitly at each stage. Stellarium’s display of weekday names provides an additional check which lets no room for speculations, it establishes that in Stellarium no other calendar convention than the astronomical year numbering is intended. And last not least, the historical transmission hypotheses are speculative. There are no direct historical records that Playfair used Cassini’s year zero or that Pingree made the proposed sign error. Despite these limitations, the result is intriguing.

7 Conclusion

The traditional Indian chronology assumes that an extraordinary planetary configuration occurred at the KY epoch. Our retrocalculation and Stellarium check identify a planetary configuration compatible with one historically attested interpretation of “Revatī” and the KY. Implementations of alternative configurations and interpretations—such as solar eclipses combined with planetary concentrations—did not find real-world events. The candidate that passes the Stellarium check is 17–18 February 3103 BCE, and Āryabhaṭa’s description of his twenty-third year may be interpreted as referring to this same year. We further argue that the conventional dates 18 February 3102 BCE and 18 February 3101 BCE warrant renewed examination, although the precise historical transmission of those dates remains to be established. We hypothesize that the apparent discrepancies discussed in modern Indology may reflect errors in Western transmission or conversion, but this hypothesis requires further archival and textual confirmation. Constructive critics are invited to calculate or demonstrate graphically another realistically possible configuration that satisfies the conditions found in existing KY traditions.

Appendix A. Python Program

To investigate a planetary two-constellation concentration between 3500 BCE and 2501 BCE, the astronomical Python package `Skyfield` is utilized. The calculations rely on the Jet Propulsion Laboratory (JPL) DE441 ephemerides, which provide high-precision positioning for ancient dates. The historical astronomical center of Ujjain, India, serves as the topocentric observation point. If one wants to include the input start date in the iteration, this input should be the day before the start date. We didn't focus on this detail, more interesting for us was how many times in a millennium such a broader planetary concentration would occur. The script filters dates based on the ecliptic longitudes of the inner planets (Mercury, Venus, Mars) and outer planets (Jupiter, Saturn) relative to the fixed star Revatī (ζ Piscium):

Listing 1: Skyfield script for calculating ancient topocentric planetary concentrations

```
# Before starting this Python program, Skyfield must be installed with
# pip install skyfield
#
# In first start you have to wait, it will install the de441.bsp data.
# To calculate candidates for the Kali Yuga epoch in a time window of 1000
# years, it may need ca. 15 to 20 minutes

from skyfield.api import Loader, Topos, Star
from skyfield import almanac
import math
load = Loader('./skyfield_data')
# DE441 is better than DE440 for ancient dates
eph = load('de441.bsp')
UJJAIN = Topos(latitude_degrees=23.17, longitude_degrees=75.79, elevation_m=494)
observer = eph['earth'] + UJJAIN
earth = eph['earth']
mercury = eph['mercury barycenter']
venus = eph['venus barycenter']
mars = eph['mars barycenter']
jupiter = eph['jupiter barycenter']
saturn = eph['saturn barycenter']
ts = load.timescale()
# Revati = Zeta Piscium (J2000 Koordinaten)
revati = Star(
    ra_hours=1 + 13/60 + 44.118/3600,
    dec_degrees=7 + 34/60 + 31.32/3600
)
# Your start and end dates (Julian-calendar, B.C.E.)
start_date = [-3500, 1, 1]
end_date = [-2501, 12, 31]
def is_julian_leap_year(y):
    """Check leap year for astronomical Julian calendar."""
    return y % 4 == 0
def days_in_julian_month(y, m):
    """Return number of days in a given Julian month."""
    if m in (1, 3, 5, 7, 8, 10, 12):
        return 31
    elif m in (4, 6, 9, 11):
        return 30
    elif m == 2:
        return 29 if is_julian_leap_year(y) else 28
def ecliptic_longitude(t, body):
    """Topocentric ecliptic longitude in degrees, epoch of date, as seen from
```

```

Ujjain. """
# Changed from earth.at(t) to observer.at(t) for topocentric calculation
astrometric = observer.at(t).observe(body).apparent()
lat, lon, dist = astrometric.ecliptic_latlon(epoch=None) # None = epoch of
date
return lon.degrees
def sep(a, b):
    """Calculate angular separation between two longitudes."""
    s = abs((a - b + 180) % 360 - 180)
    return s
print("-" * 100)
y, m, d = start_date
end_y, end_m, end_d = end_date
# Use a while loop to guarantee termination exactly when the end date is
reached/passed
while (y, m, d) <= (end_y, end_m, end_d):
    # Increment to the next Julian calendar day
    d += 1
    if d > days_in_julian_month(y, m):
        d = 1
        m += 1
        if m > 12:
            m = 1
            y += 1
    # Stop immediately if the incremented date exceeds the end date
    if (y, m, d) > (end_y, end_m, end_d):
        break
    t = ts.ut1(y, m, d, 1, 0, 0) # between midnight and sunrise UT on that
    Julian day
    # Now we calculate the separation of the planets. The outer planets are the
    slowest,
    # their separation doesn't need to be as close as of the inner planets,
    since according
    # to Abu Mashar the planetary concentration didn't occur completely in
    Pisces but spanned two zodiac signs.
    # We are stricter with the faster inner planets.
    L_mer = ecliptic_longitude(t, mercury)
    L_ven = ecliptic_longitude(t, venus)
    s_ven_mer = sep(L_ven, L_mer) # the separations of planets nearest to the
    sun should be the lowest
    if (s_ven_mer > 3.0): continue
    L_revati = ecliptic_longitude(t, revati) # before the outer planets, we
    consider inner planets vs Revati
    s_mer_rev = sep(L_mer, L_revati)
    if (s_mer_rev > 7.0): continue
    s_ven_rev = sep(L_ven, L_revati)
    if (s_ven_rev > 7.0): continue
    L_mars = ecliptic_longitude(t, mars) # then we can relax the strictness
    of the inner planet degrees
    s_mar_mer = sep(L_mars, L_mer)
    if (s_mar_mer > 7.0): continue
    s_mar_rev = sep(L_mars, L_revati)
    if (s_mar_rev > 7.0): continue
    s_mar_ven = sep(L_mars, L_ven)
    if (s_mar_ven > 7.0): continue
    # outer planets may span the length of two zodiac signs, 60 Degrees
    L_jup = ecliptic_longitude(t, jupiter)
    s_jup_rev = sep(L_jup, L_revati)

```

```

if (s_jup_rev > 60.0): continue
s_jup_mer = sep(L_jup, L_mer)
if (s_jup_mer > 60.0): continue
s_jup_ven = sep(L_jup, L_ven)
if (s_jup_ven > 60.0): continue
s_jup_mar = sep(L_jup, L_mars)
if (s_jup_mar > 60.0): continue
L_sat = ecliptic_longitude(t, saturn)
s_sat_rev = sep(L_sat, L_revati)
if (s_sat_rev > 60.0): continue
s_sat_mer = sep(L_sat, L_mer)
if (s_sat_mer > 60.0): continue
s_sat_ven = sep(L_sat, L_ven)
if (s_sat_ven > 60.0): continue
s_sat_mar = sep(L_sat, L_mars)
if (s_sat_mar > 60.0): continue
s_sat_jup = sep(L_jup, L_sat)
if (s_sat_jup > 60.0): continue
# Filter: Do print a date that has survived all this conditions
print(f"{y:>5}-{m:02d}-{d:02d}  ")

```

Appendix B: Wolfram Mathematica Replication Script

We try two different algorithms executed in Wolfram Mathematica to scan for broader planetary concentrations near ζ Piscium (*Revati*) between 3500 BCE and 2500 BCE. Both of them establish the astronomical validity of the 3103 BCE timeframe. Algorithm A: Using the ecliptical longitude, this one follows 1:1 Python script.

```

(* ===Standort Ujjain setzen===*)SetLocation[GeoLongitude->75.79*Degree,
GeoLatitude->23.17*Degree,GeoAltitude->0.494*KiloMeter,TimeZone->5];
(* ===Ecliptical length of an object in degrees (0,360)=== *)
(* For planets:topocentric ecliptical coordinates *)
EclipticLon[object_,day_]:=Module[{coords,dx,dy},coords=Coordinates[object,
day,ViewPoint->TopoCentric];
dx=coords[[1]];
dy=coords[[2]];
Mod[ArcTan[dx,dy]/Degree,360]];
(* For the star Revati=Zeta Piscium: *)
(* EquatorCoordinates returns Ascension/Declination, then conversion in
ecliptic *)
EclipticLonStar[day_]:=Module[{asc,dec,tilt,c,s,cosD,sinA,cosA,sinD,x,y,z,
lon},{asc,dec}={Ascension,Declination}/.
EquatorCoordinates[Zeta.Pisces,day,ViewPoint->Sun];
asc=N[asc];dec=N[dec];
tilt=EarthTilt[day]*N[Degree];
c=Cos[tilt];s=Sin[tilt];
cosD=Cos[dec];sinD=Sin[dec];
cosA=Cos[asc];sinA=Sin[asc];
(* Conversion equatorial->ecliptical *)
x=cosA*cosD;
y=c*sinA*cosD+s*sinD;
z=-s*sinA*cosD+c*sinD;
lon=ArcTan[x,y]/Degree;
Mod[lon,360]];
(* === Separation in ecliptical length, identical with Python sep()=== *)
sepLon[a_,b_]:=Abs[Mod[a-b+180,360]-180];
FixedToMJD[fixed_]:=fixed-1721424.5+2415020.0 ;

```

```

(* MJD 0=17. Nov 1858; Fixed 0=1. Jan 1 CE (Gregorian)*)
(* Alternative:directly Julian->Gregorian->ModifiedJulianDay. In the 19th
   century there were many ideas to calculate MJD, including such ones
   where the epoch did not start on 1. Januar. In our case we have to apply
   a correction of 339 days *)
JulianToMJD[jdate_List]:=Module[{fixed,greg},fixed=ToFixed[Apply[Julian,
   jdate]];
greg=Gregorian[fixed];
ModifiedJulianDay[{CYear[greg],CMonth[greg],CDay[greg],1,0,0}]];
(* === Central program ===*)
Module[{start,end,akt,y,m,d,day,LMer,LVen,LMars,LJup,LSat,LRev,sVenMer,
   sMerRev,sVenRev,sMarMer,sMarRev,sMmarVen,sJupRev,sJupMer,sJupVen,sJupMar,
   sSatRev,sSatMer,sSatVen,sSatMar,sSatJup},start=ToFixed[Julian
   [-3499,1,1]];
end=ToFixed[Julian[-2500,12,31]];
(* We check the millenium 3500 BCE to 2501 BCE *)
Do[(* Extract Julian date *)
{y,m,d}=Julian[akt]/. Julian->List;
(* Modified Julian Day for Planetarium incl. correction to 1. Jan *)
day=JulianToMJD[{y,m,d}]-339;
(* Calculate ecliptical lengths *)
LMer=EclipticLon[Mercury,day];
LVen=EclipticLon[Venus,day];
(* Venus-Mercury:strictest filter *)
sVenMer=sepLon[LVen,LMer];
If[sVenMer>3.0,Continue[]];
LRev=EclipticLonStar[day];
sMerRev=sepLon[LMer,LRev];
If[sMerRev>7.0,Continue[]];
sVenRev=sepLon[LVen,LRev];
If[sVenRev>7.0,Continue[]];
LMars=EclipticLon[Mars,day];
sMarMer=sepLon[LMars,LMer];
If[sMarMer>7.0,Continue[]];
sMarRev=sepLon[LMars,LRev];
If[sMarRev>7.0,Continue[]];
sMarVen=sepLon[LMars,LVen];
If[sMarVen>7.0,Continue[]];
(* Outer planets:60 degree tolerance *)
LJup=EclipticLon[Jupiter,day];
sJupRev=sepLon[LJup,LRev];
If[sJupRev>60.0,Continue[]];
sJupMer=sepLon[LJup,LMer];
If[sJupMer>60.0,Continue[]];
sJupVen=sepLon[LJup,LVen];
If[sJupVen>60.0,Continue[]];
sJupMar=sepLon[LJup,LMars];
If[sJupMar>60.0,Continue[]];
LSat=EclipticLon[Saturn,day];
sSatRev=sepLon[LSat,LRev];
If[sSatRev>60.0,Continue[]];
sSatMer=sepLon[LSat,LMer];
If[sSatMer>60.0,Continue[]];
sSatVen=sepLon[LSat,LVen];
If[sSatVen>60.0,Continue[]];
sSatMar=sepLon[LSat,LMars];
If[sSatMar>60.0,Continue[]];
sSatJup=sepLon[LJup,LSat];

```

```

If[sSatJup>60.0,Continue[]];
(* All conditions satisfied-print date *)
Print[Julian[akt]],{akt,start,end}]]
(* Output *)
17 February 3103 B.C.E.
18 February 3103 B.C.E.

```

Appendix C: Second Wolfram Mathematica Replication Script

Algorithm B: Using the function Separation from the Planetarium package.

```

Separation[object1_,object2_,zeit_,ort_]:= (Geo$Latitude=ort[[2]];
Geo$Longitude=ort[[3]];
Geo$Altitude=ort[[4]]/1000;
GMT$diff=ort[[5]];
Separation[object1,object2,zeit]);
Ujjain[]=Location["Ujjain, India",Deg[23.17],Deg[75.79],Mt[494],5];
(* <<"c:\\Stars250.m"; *)
Module[{start, end, akt,afm,hs,s1,s2,s3,s4, s5, s6, s7, s8, s9, s10, s11,
s12, s13, s14, s15},
start=ToFixed[Julian[-3500,1,1]]; end=ToFixed[Julian[-2501,12,31]];
Do[afm=Julian[akt]/. Julian -> List;
(* Separation doesn't output in Degrees since it considers also latitudes
as well as longitudes.
The search strategy remains: we expect the closest planetary concentrations
of the fast planets near the sun and then a broader planetary
concentration of the outer planets which may span two zodiac signs *)
s1= Separation[Mercury,Zeta. Pisces,afm,Ujjain[]];If[s1 > 0.05,Continue[]];
s2=Separation[Venus,Zeta. Pisces,afm,Ujjain[]];If[s2 > 0.05, Continue[]];
s3=Separation[Mercury,Venus,afm,Ujjain[]];If[s3 > 0.1,Continue[]];
(* the last inner planet *)
s4=Separation[Mars,Zeta. Pisces,afm,Ujjain[]];If[s4 > 0.1,Continue[]];
s5=Separation[Mercury,Mars,afm,Ujjain[]];If[s5 > 0.2,Continue[]];
s6=Separation[Venus,Mars,afm,Ujjain[]];If[s6 > 0.2,Continue[]];
(* And now the outer planets *)
s7=Separation[Jupiter,Zeta. Pisces,afm,Ujjain[]];If[s7 > 1.0,Continue[]];s8
=Separation[Saturn,Zeta. Pisces,afm,Ujjain[]];If[s8 > 1.0,Continue[]];
s9=Separation[Mercury,Jupiter,afm,Ujjain[]];If[s9 > 1.0,Continue[]];
s10=Separation[Venus,Jupiter,afm,Ujjain[]];If[s10 > 1.0, Continue[]];
s11=Separation[Mars,Jupiter,afm,Ujjain[]];If[s11 > 1.0,Continue[]];
s12= Separation[Mercury,Saturn,afm,Ujjain[]];If[s12 > 1.0, Continue[]];
s13=Separation[Venus,Saturn,afm,Ujjain[]]; If[s13 > 1.0, Continue[]];
s14=Separation[Mars,Saturn,afm,Ujjain[]]; If[s14 > 1.0, Continue[]];
s15=Separation[Jupiter,Saturn,afm,Ujjain[]]; If[s15 > 1.0,Continue[]];
Print[Julian[akt]], {akt,start,end}]]
(* Output: *)
(* 17 February 3103 B.C.E. *)
(* 18 February 3103 B.C.E. *)

```

Further mathematical verification of the 1-year calendar shift and the matching *Brihaspatvara* (Thursday) weekday signature via the native calendar conversion:

```

ConvertDateTo[Julian[-3103, 2, 18], OldHinduSolar] // FullForm
(* Output: OldHinduSolar[-1, 1, 1] -> Brihaspatvara, 1 Mesha -1 K.Y. *)
(* The official, shifted calendar epoch: *)
OldHinduSolar
(* Output: Sukravara, 1 Mesha 0 K.Y. *)

```

Appendix D: hypothese.py

```
from skyfield.api import Loader, Topos, Star
from skyfield import almanac
import math
load = Loader('./skyfield_data')
# DE441 is better than DE440 for ancient dates
eph = load('de441.bsp')
UJJAIN = Topos(latitude_degrees=23.17, longitude_degrees=75.79, elevation_m=494)
observer = eph['earth'] + UJJAIN
earth = eph['earth']
mercury = eph['mercury barycenter']
venus = eph['venus barycenter']
mars = eph['mars barycenter']
jupiter = eph['jupiter barycenter']
saturn = eph['saturn barycenter']
ts = load.timescale()
# Your start and end dates (Julian-calendar, B.C.E.)
start_date = [-3103, 2, 17]
end_date = [-3103, 2, 18]
def is_julian_leap_year(y):
    """Check leap year for astronomical Julian calendar."""
    return y % 4 == 0
def days_in_julian_month(y, m):
    """Return number of days in a given Julian month."""
    if m in (1, 3, 5, 7, 8, 10, 12):
        return 31
    elif m in (4, 6, 9, 11):
        return 30
    elif m == 2:
        return 29 if is_julian_leap_year(y) else 28
def ecliptic_longitude(t, body):
    """Topocentric ecliptic longitude in degrees, epoch of date, as seen from Ujjain."""
    # Changed from earth.at(t) to observer.at(t) for topocentric calculation
    astrometric = observer.at(t).observe(body).apparent()
    lat, lon, dist = astrometric.ecliptic_latlon(epoch=None) # None = epoch of date
    return lon.degrees
def sep(a, b):
    """Calculate angular separation between two longitudes."""
    s = abs((a - b + 180) % 360 - 180)
    return s
# Updated header to correctly match the printed variables below
print("-" * 100)
y, m, d = start_date
end_y, end_m, end_d = end_date
while True:
    t = ts.ut1(y, m, d, 12, 0, 0) # noon UT on that Julian day
    L_jup = ecliptic_longitude(t, jupiter)
    L_sat = ecliptic_longitude(t, saturn)
    L_mars = ecliptic_longitude(t, mars)
    L_ven = ecliptic_longitude(t, venus)
    L_mer = ecliptic_longitude(t, mercury)
    # calculation of the separations
    s_ven_mer = sep(L_ven, L_mer)
    s_mar_mer = sep(L_mars, L_mer)
```

```

s_mar_ven = sep(L_mars, L_ven)
s_jup_mer = sep(L_jup, L_mer)
s_jup_ven = sep(L_jup, L_ven)
s_jup_mar = sep(L_jup, L_mars)
s_sat_mer = sep(L_sat, L_mer)
s_sat_ven = sep(L_sat, L_ven)
s_sat_mar = sep(L_sat, L_mars)
s_jup_sat = sep(L_jup, L_sat)
# Do print if all of the specific conditions are met
if True:
print(f"{y:>5}-{m:02d}-{d:02d}  "
f"{s_ven_mer:8.2f}  "
f"{s_mar_mer:8.2f}  "
f"{s_mar_ven:8.2f}  "
f"{s_jup_mer:8.2f}  "
f"{s_jup_ven:8.2f}  "
f"{s_jup_mar:8.2f}  "
f"{s_sat_mer:8.2f}  "
f"{s_sat_ven:8.2f}  "
f"{s_sat_mar:8.2f}  "
f"{s_jup_sat:8.2f}  ")
# Stop if we have processed the end date
if (y, m, d) == (end_y, end_m, end_d):
break
# Increment to the next Julian calendar day
d += 1
if d > days_in_julian_month(y, m):
d = 1
m += 1
if m > 12:
m = 1
y += 1

```

Appendix E: gegenhypotheseA.py

```

from skyfield.api import Loader, Topos, Star
from skyfield import almanac
import math
load = Loader('./skyfield_data')
# DE441 is better than DE440 for ancient dates
eph = load('de441.bsp')
UJJAIN = Topos(latitude_degrees=23.17, longitude_degrees=75.79, elevation_m=494)
observer = eph['earth'] + UJJAIN
earth = eph['earth']
mercury = eph['mercury barycenter']
venus = eph['venus barycenter']
mars = eph['mars barycenter']
jupiter = eph['jupiter barycenter']
saturn = eph['saturn barycenter']
ts = load.timescale()
# Your start and end dates (Julian-calendar, B.C.E.)
start_date = [-3102, 2, 17]
end_date = [-3102, 2, 18]
def is_julian_leap_year(y):
"""Check leap year for astronomical Julian calendar."""

```

```

return y % 4 == 0
def days_in_julian_month(y, m):
    """Return number of days in a given Julian month."""
    if m in (1, 3, 5, 7, 8, 10, 12):
        return 31
    elif m in (4, 6, 9, 11):
        return 30
    elif m == 2:
        return 29 if is_julian_leap_year(y) else 28
    def ecliptic_longitude(t, body):
        """Topocentric ecliptic longitude in degrees, epoch of date, as seen from
        Ujjain."""
        # Changed from earth.at(t) to observer.at(t) for topocentric calculation
        astrometric = observer.at(t).observe(body).apparent()
        lat, lon, dist = astrometric.ecliptic_latlon(epoch=None) # None = epoch of
        date
        return lon.degrees
    def sep(a, b):
        """Calculate angular separation between two longitudes."""
        s = abs((a - b + 180) % 360 - 180)
        return s
    # Updated header to correctly match the printed variables below
    print("-" * 100)
    y, m, d = start_date
    end_y, end_m, end_d = end_date
    while True:
        t = ts.ut1(y, m, d, 12, 0, 0) # noon UT on that Julian day
        L_jup = ecliptic_longitude(t, jupiter)
        L_sat = ecliptic_longitude(t, saturn)
        L_mars = ecliptic_longitude(t, mars)
        L_ven = ecliptic_longitude(t, venus)
        L_mer = ecliptic_longitude(t, mercury)
        # calculation of the separations
        s_ven_mer = sep(L_ven, L_mer)
        s_mar_mer = sep(L_mars, L_mer)
        s_mar_ven = sep(L_mars, L_ven)
        s_jup_mer = sep(L_jup, L_mer)
        s_jup_ven = sep(L_jup, L_ven)
        s_jup_mar = sep(L_jup, L_mars)
        s_sat_mer = sep(L_sat, L_mer)
        s_sat_ven = sep(L_sat, L_ven)
        s_sat_mar = sep(L_sat, L_mars)
        s_jup_sat = sep(L_jup, L_sat)
        # Do print if all of the specific conditions are met
        if True:
            print(f"{y:>5}-{m:02d}-{d:02d}  "
                  f"{s_ven_mer:8.2f}  "
                  f"{s_mar_mer:8.2f}  "
                  f"{s_mar_ven:8.2f}  "
                  f"{s_jup_mer:8.2f}  "
                  f"{s_jup_ven:8.2f}  "
                  f"{s_jup_mar:8.2f}  "
                  f"{s_sat_mer:8.2f}  "
                  f"{s_sat_ven:8.2f}  "
                  f"{s_sat_mar:8.2f}  "
                  f"{s_jup_sat:8.2f}  ")
        # Stop if we have processed the end date
        if (y, m, d) == (end_y, end_m, end_d):

```

```

break
# Increment to the next Julian calendar day
d += 1
if d > days_in_julian_month(y, m):
d = 1
m += 1
if m > 12:
m = 1
y += 1

```

Appendix F: gegenhypotheseB.py

```

from skyfield.api import Loader, Topos, Star
from skyfield import almanac
import math
load = Loader('./skyfield_data')
# DE441 is better than DE440 for ancient dates
eph = load('de441.bsp')
UJJAIN = Topos(latitude_degrees=23.17, longitude_degrees=75.79, elevation_m=494)
observer = eph['earth'] + UJJAIN
earth = eph['earth']
mercury = eph['mercury barycenter']
venus = eph['venus barycenter']
mars = eph['mars barycenter']
jupiter = eph['jupiter barycenter']
saturn = eph['saturn barycenter']
ts = load.timescale()
# Your start and end dates (Julian-calendar, B.C.E.)
start_date = [-3101, 2, 17]
end_date = [-3101, 2, 18]
def is_julian_leap_year(y):
    """Check leap year for astronomical Julian calendar."""
    return y % 4 == 0
def days_in_julian_month(y, m):
    """Return number of days in a given Julian month."""
    if m in (1, 3, 5, 7, 8, 10, 12):
        return 31
    elif m in (4, 6, 9, 11):
        return 30
    elif m == 2:
        return 29 if is_julian_leap_year(y) else 28
def ecliptic_longitude(t, body):
    """Topocentric ecliptic longitude in degrees, epoch of date, as seen from Ujjain."""
    # Changed from earth.at(t) to observer.at(t) for topocentric calculation
    astrometric = observer.at(t).observe(body).apparent()
    lat, lon, dist = astrometric.ecliptic_latlon(epoch=None) # None = epoch of date
    return lon.degrees
def sep(a, b):
    """Calculate angular separation between two longitudes."""
    s = abs((a - b + 180) % 360 - 180)
    return s
# Updated header to correctly match the printed variables below
print("-" * 100)

```

```

y, m, d = start_date
end_y, end_m, end_d = end_date
while True:
    t = ts.ut1(y, m, d, 12, 0, 0) # noon UT on that Julian day
    L_jup    = ecliptic_longitude(t, jupiter)
    L_sat    = ecliptic_longitude(t, saturn)
    L_mars    = ecliptic_longitude(t, mars)
    L_ven    = ecliptic_longitude(t, venus)
    L_mer    = ecliptic_longitude(t, mercury)
    # calculation of the separations
    s_ven_mer = sep(L_ven, L_mer)
    s_mar_mer = sep(L_mars, L_mer)
    s_mar_ven = sep(L_mars, L_ven)
    s_jup_mer = sep(L_jup, L_mer)
    s_jup_ven = sep(L_jup, L_ven)
    s_jup_mar = sep(L_jup, L_mars)
    s_sat_mer = sep(L_sat, L_mer)
    s_sat_ven = sep(L_sat, L_ven)
    s_sat_mar = sep(L_sat, L_mars)
    s_jup_sat = sep(L_jup, L_sat)
    # Do print if all of the specific conditions are met
    if True:
        print(f"{y:>5}-{m:02d}-{d:02d}  "
              f"{s_ven_mer:8.2f}  "
              f"{s_mar_mer:8.2f}  "
              f"{s_mar_ven:8.2f}  "
              f"{s_jup_mer:8.2f}  "
              f"{s_jup_ven:8.2f}  "
              f"{s_jup_mar:8.2f}  "
              f"{s_sat_mer:8.2f}  "
              f"{s_sat_ven:8.2f}  "
              f"{s_sat_mar:8.2f}  "
              f"{s_jup_sat:8.2f}  ")
    # Stop if we have processed the end date
    if (y, m, d) == (end_y, end_m, end_d):
        break
    # Increment to the next Julian calendar day
    d += 1
    if d > days_in_julian_month(y, m):
        d = 1
        m += 1
    if m > 12:
        m = 1
        y += 1

```

References

- [1] Jean-Sylvain Bailly. *Traité de l'astronomie indienne et orientale*. Debure, Paris, 1787.
- [2] Roger Billard. *L'astronomie indienne: Investigation sur l'histoire astronomique des textes sanskrits*. École Française d'Extrême-Orient, Paris, 1971.
- [3] Al-Bīrūnī. *The Chronology of Ancient Nations*. Translated and edited by C. Edward Sachau. London: W. H. Allen, 1879. (Contains al-Bīrūnī's report of Abū Maʿshar's statement on the Deluge planetary concentration.)
- [4] Jacques Cassini. *Tables astronomiques du soleil, de la lune, des planetes, des etoiles fixes, et des satellites de Jupiter et de Saturne*. Imprimerie Royale, Paris, 1740.
- [5] Guillaume Le Gentil. *Voyage dans les mers de l'Inde, fait par ordre du Roi, à l'occasion du passage de Vénus sur le disque du soleil le 6 juin 1761 & le 3 du même mois 1769*. Imprimerie Royale, Paris, 1779.
- [6] Abū Maʿshar al-Balkhī (Albumasar). *De magnis coniunctionibus annorum revolutionibus ac eorum profectionibus*. Venice, 1515. (Latin translation of *Kitāb al-qirānāt* by Joannes Hispalensis; contains the Flood planetary concentration in Virgo and the 3950-year interval to the concentration in Scorpio.)
- [7] Abu Maʿshar al-Balkhi (Albumasar). *De magnis coniunctionibus* (*Kitāb al-qirānāt*). Bagdad, c. 886 CE (Latin translation by Johannes Hispalensis, 1184).
- [8] Al-Maqrīzī. *al-Ḥabar ʿan al-baṣār*. 15th century CE. (Historical compilation containing the citation of Abū Maʿshar on the Deluge planetary concentration.)
- [9] Raymond Mercier. The Meridian of Gwalior. *History of Oriental Astronomy*, pages 165–182. Cambridge University Press, 1987.
- [10] David Pingree. Astronomy and Astrology in India and Iran. *Isis*, 54(2):229–246, 1963. (Discusses Abū Maʿshar's date for the Flood, the planetary concentration of -3101, and its derivation from Hindu sources.)
- [11] David Pingree. *The Thousands of Abū Maʿshar*. Studies of the Warburg Institute, 30. London, 1968. (Discusses Abū Maʿshar's use of Māshā'allāh's chronological scheme and the Flood planetary concentration.)
- [12] David Pingree. *Jyotiṣśāstra: Astral and Mathematical Literature*. Otto Harrassowitz, Wiesbaden, 1978.
- [13] John Playfair. Remarks on the Astronomy of the Brahmins. *Transactions of the Royal Society of Edinburgh*, 2(2):135–192, 1790.
- [14] K. S. Raghavan. *The Date of the Mahabharata War*. Srinivasanagar, 1969.
- [15] B. N. Narahari Achar, "Astronomy of the Mahabharata War," *Journal of the Indian Council of Philosophical Research*, vol. 23, no. 4, pp. 1–28, 2006.
- [16] S. Kak. On the Chronology of Ancient India. *Indian Journal of History of Science*, 22, 1987.
- [17] S. Kak. Knowledge of Planets in the Third Millennium BC. *Quarterly Journal of the Royal Astronomical Society*, 37, 1996.

- [18] S. Kak, *The Astronomical Code of the Ṛgveda*, 3rd ed., Oklahoma State University / Academia, 2016.
- [19] *Sūryasiddhānta*. Madhyamādhikāra, 1.57.
- [20] Brahmagupta. *Brāhmasphuṭasiddhānta*. Madhyamādhikāra and associated commentary.
- [21] C. Edward Sachau. *Alberuni's India*, vol. 1. London: Kegan Paul, Trench, Trübner & Co., 1910. (Contains al-Bīrūnī's summary of Abū Ma'shar's theory that the Deluge occurs at the concentration of the planets at the end of each *caturyuga* and the beginning of each *kaliyuga*.)
- [22] Wolfram Research, Inc. *Mathematica*, Version 12.0. Wolfram Research, Inc., Champaign, IL, 2019.
- [23] Stellarium Software. *Stellarium*, version 26.2, 2026. <https://stellarium.org/>.
- [24] Park, R. S., Folkner, W. M., Williams, J. G., & Boggs, D. H. (2021). The JPL Planetary and Lunar Ephemerides DE440 and DE441. *The Astronomical Journal*, 161(3), 105. <https://doi.org/10.3847/1538-3881/abd414>
- [25] B. Rhodes. Skyfield: Elegant Astronomy for Python, 2026. <https://rhodesmill.org/skyfield/>. B. Rhodes, "Skyfield: Elegant Astronomy for Python," GitHub Repository, 2026. <https://github.com> [Accessed: September 29, 2026]